M1 - Concept Proposal Documentation

SI 612 - Section 001
Team 3
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Introduction

Pervasive Interaction Design focuses on embedding computation and interaction into everyday environments rather than relying solely on traditional screen-based interfaces. In our understanding, this approach explores how sensors, connected objects, and ambient systems can support human activities in subtle and contextual ways. Instead of designing standalone apps or devices, we view pervasive interaction as designing ecosystems where physical spaces, smart objects, and users work together, allowing technology to become an invisible but meaningful part of daily routines.

In Milestone 1, our team focused on early-stage exploration, ideation, and opportunity framing. We began by individually generating over 80 ideas across diverse environments and user groups using methods such as a 10x10 matrix, Google and IDEO design kits, and lateral thinking. We then synthesized these ideas collaboratively through affinity diagramming and dot voting in Figjam to identify shared interests and promising contexts. This process helped us move from broad and speculative concepts toward more focused environments, ultimately narrowing our scope to Healthcare and Smart Everyday Objects. M1 also introduced us to contextual inquiry, which we used to start grounding our ideas in real-world behaviors rather than assumptions.

The primary goal of M1 was to move from divergent brainstorming to a clearly defined opportunity space that balances creativity, feasibility, and user relevance. Rather than producing a finalized product, we aimed to identify realistic environments and audiences for deeper research while reflecting on recruitability and practical constraints. By the end of this milestone, we sought to establish a shared direction for Milestone 2, supported by structured ideation, team alignment, and an initial understanding of user needs that will guide our upcoming field research and concept development.

Ideation Process

During the ideation process, each member of our team individually came up with 20+ initial ideas, leading to over 80 ideas altogether. These ideas spanned across several diverse environments

such as grocery stores, laundry rooms, and corporate offices, with different audiences in mind ranging from graduate students to senior citizens.

Opportunities

- AUTOMATE A: Have a shopping cart robot to automatically follow your back when shopping
- AUGMENT B: Smart shopping cart that allows users to search for specific products and navigate them to where the product is located
- AUTOMATE C: Product automatically checks out as users put them in the cart, and automatically pays as the user walks out the store
- AUTOMATE D: Robot that transports groceries to the user's car

- AUGMENT E&F: Smart home system that manages clocks, alarms and other devices (smart curtains, etc) collectively and setup wake-up time for everyone
- AUGMENT G: Smart bed that use various ways (visual, auditory, physical, etc.) to wake children up if they stayed in bed for too long
- AUTOMATE H: Smart school bus that self-drives and picks up children from different locations along best route

- AUGMENT I: Provide task list & summary of the day verbally while on the way to the office
- AUTOMATE J: Smart device that records, take notes, and generate summaries during the meeting
- AUGMENT K: Ambient lighting & sound on workstation that help workers change mood and/or stay focused

- AUGMENT L: Smart home system that can pull up daily agenda in different places/devices at home
- AUGMENT M: Smart home system that can pull up display/camera and join zoom meetings anywhere in at home
- AUGMENT N: Smart home office system that allows users to work on anywhere at home (maybe with AR/VR)

- AUTOMATE O: Healthcare robot that will auto-detect the user's health conditions and remind them when there's an alarm
- AUTOMATE P: At-home robot personal doctor that helps provide medication whenever needed
- AUTOMATE Q: Drone delivery that delivers medication in minutes in case of emergency
- AUGMENT R: Smart bed that automatically adjusts shape and lighting to make the user comfortable according to body conditions

- AUGMENT T: An all-in-one social platform that shares everyone's info. With AR glasses, it highlights people with similar interests with you and encourage you to start a conversation with them.
- AUGMENT U: Detects the atmosphere (or chemistry) between you and the other person, and reminds you what to do next (what direction of the conversation to go to or whether you should continue with the conversation)
- AUTOMATE V: Automatically record everyone that you had conversation with today, and you can chat or hang out with them in the future

Figure 1: Visual of Ideas by Team Member Andy Xu

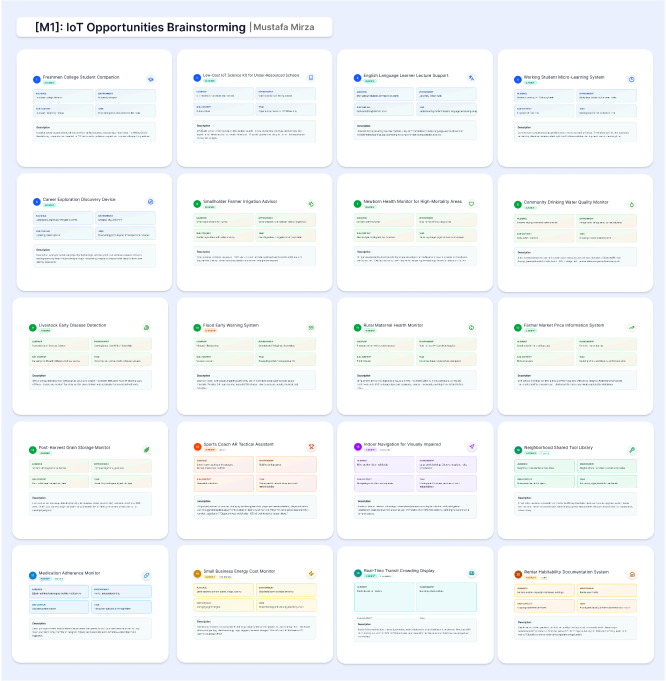

Figure 2: Visual of Ideas by Team Member Mustafa Mirza

To come up with these ideas, members of our team utilized a 10x10 matrix, design kits from Google and IDEO, and lateral thinking. Using a 10x10 matrix allowed us to come up with 10 sensors, 10 items, and finally 100 smart items that fell at the intersections of those sensors and

items. The design kits from Google and IDEO were helpful during ideation because they encouraged us to approach brainstorming from a “How Might We” perspective when thinking about daily routines and where smart objects might help everyday tasks. Using lateral thinking pushed us to break free from assumptions and approach brainstorming in a fun way by exploring “impossible” or ridiculous solutions. Each of these was useful to our members, but lateral thinking was the most useful to our team. As a result of using a lateral thinking method during ideation, our team had many creative and unconventional ideas that were interesting to explore, such as a “shopping cart robot that follows you around the store”. Lateral thinking was useful to our team because it allowed us to brainstorm and ideate in an unfiltered way.

Selection Process

To evaluate opportunities and select environments for contextual inquiry, our team began by collaborating in a Figjam file and sharing our individual ideas. We affinity diagrammed to create distinct environment categories. We organized them on sticky notes and as a group, chose the environments that seemed to have the most overlap between members.

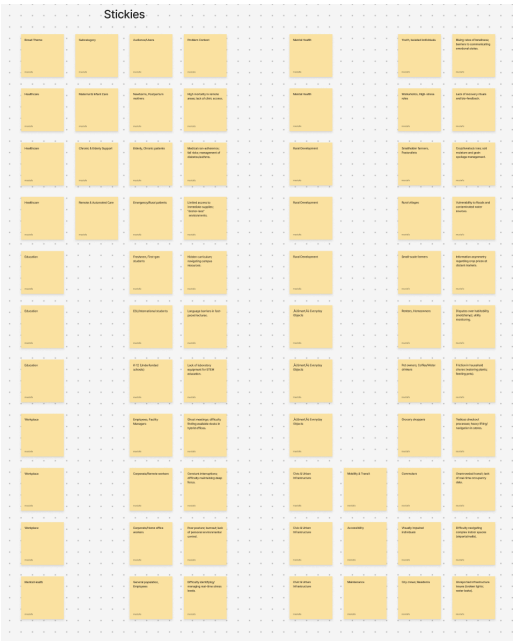

Figure 3: Sticky Notes in Figjam

The environment/context categories that our members had focused on the most during ideation were Healthcare, Education, Rural Development, Smart Everyday Objects, Workplace, and Mental Health/Self Care. Once we identified these, we utilized dot voting. Each member had three dot votes that they could use to select the environment or context categories they were most interested in. After dot voting, Healthcare and Smart Everyday Objects were selected as the environments/contexts that we would explore for contextual inquiry.

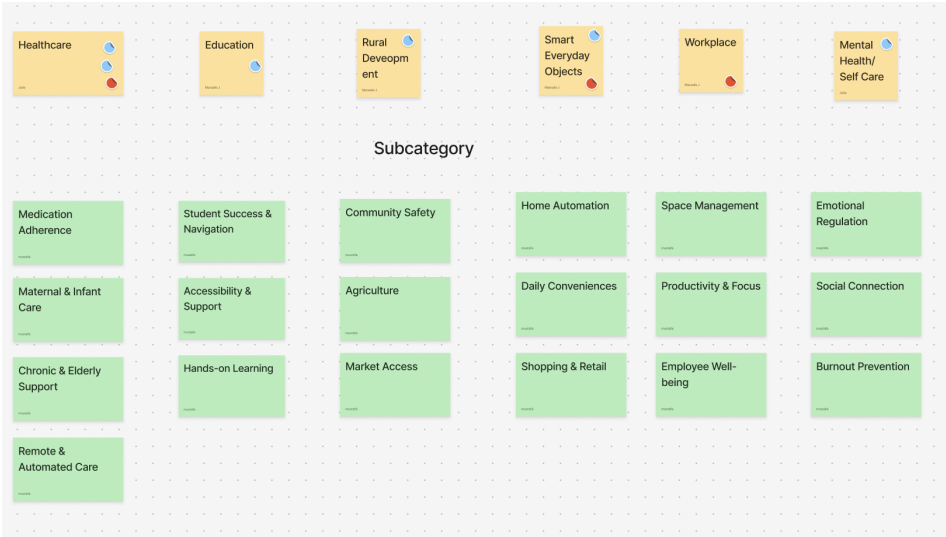

Figure 4: Dot-Voting Method in Figjam

Dot-voting was the most useful elimination/selection method used, because it ensured a democratic and fair “majority rule”. Using this method helped our team move from agonizing over six diverse categories to having a clear and popular direction that everyone could get behind. It made elimination and selection faster and more efficient.

Contextual Inquiry Overview

Contextual Inquiry is a user-centered design research method where the design team observes users in their natural environment as they perform their tasks. Its key elements include context (going to where the work happens), partnership (engaging with the user as an expert in their own life), interpretation (validating observations with the user), and focus (guiding the inquiry based on project goals).

At this stage of M1, Contextual Inquiry is critical because it helps us move beyond our initial assumptions like "users forget because they are lazy" to understanding the systemic reality or “users forget because their environment is chaotic". It reveals the "invisible" workarounds and physical barriers that surveys or interviews often miss.

Research Type: Qualitative Research. Ideal Participants: 3-5 users per environment to observe key behaviors.

Environment A: Student Housing & Personal Artifacts

Participants:

- Count: 3 participants (Team members & Roommates).

- Recruiting Criteria: We sought participants who were individuals aged 20-25 managing daily supplements or prescription medications while balancing irregular class schedules.
- Recruiting Process: Since our team fits the target demographic, we analyzed our own bags and recruited live-in roommates. Consent was obtained verbally for photographing personal items.
- Characteristics: Participants were graduate and undergraduate students who carry bags to campus daily.

Logistics:

- Location: Private apartments and places on campus where bags are packed and or unpacked.
- User Tasks: We focused on the task of how students organize their daily supplies. We analyzed their backpack contents once dumped out to see where health items physically lived.
- Interaction: Sessions involved dumping the entire contents of a backpack onto a table and mapping the stratification of items, for example what is on top vs. what is at the bottom. Interactions lasted approximately 20 minutes.

Results and Top 3 Themes:

1. Finding: In our backpack audits, we found that pain relievers and emergency meds were consistently crushed at the bottom of bags or hidden in obscure, zipped pockets. They were "invisible" during moments of need.
    - Evidence: Photos of chaotic bag interiors where a blister pack of ibuprofen was indistinguishable from old receipts and charging cables.
2. Finding: Tech items like laptops and chargers, and sustenance items like water and snacks always occupied the easy access pockets. Health items were relegated to deep storage, indicating they are not prioritized in the user's mental model of getting ready.
    - Evidence: Diagram of a backpack showing the tech zone that was more accessible vs. the health zone, which was buried away.
3. Finding: Most meds found in bags were for emergency use like headaches and allergies, not daily use. This suggests students don't trust their bags for daily adherence, creating a gap for on-the-go management.
    - Evidence: Participant quote: "I only keep this here for emergencies; if I need to take a pill every day, I leave it at home so I don't lose it."

Pros/Cons of this Opportunity Space:

- Pros: This environment offers high accessibility for testing and high impact. We can easily iterate on prototypes in our own homes, and the problem (adherence) has a direct link to student well-being.

- Cons: Privacy is a significant challenge; capturing data about personal medications requires careful ethical handling. Additionally, "morning routines" are deeply ingrained and hard to change, meaning our intervention must be very low-friction.

Environment B: Campus & Fitness Spaces + Libraries

Participants:

- Count: approx 10 participants (Passive observation).
- Recruiting Criteria: We looked for the general student population settling into new spaces and preparation behaviors in the gym context.
- Recruiting Process: Public Observation. We stationed team members in university libraries and gym locker rooms. No direct interaction/consent was required as observations were anonymous and in public settings.
- Characteristics: Students are acting in certain ways, a public mode and managing social perceptions while trying to be productive or active.

Logistics:

- Location: University Libraries (Duderstadt and Hatcher) and Locker Rooms of the IMSB gym.
- User Tasks: The ritual of "settling in" at a desk and their workout preparation.
- Interaction: 10-15 minute passive observation sessions, taking notes on what objects were removed from bags and in what order.

Results (Top 3 Themes):

1. Finding: When students sit down at a library, they immediately unpack a specific "micro-ecosystem": laptop, phone, water bottle. However, medication is rarely part of this setup, meaning it is not accessible during the bulk of the day.
    - Evidence: Observation logs showing 9/10 students unpack water, but 0/10 unpack health items.
2. Finding: In the gym, we observed students proudly displaying protein shakers and pre-workout (performance health). In contrast, daily medication (maintenance health) was observed to be taken furtively or hidden.
    - Evidence: Observation note: "Gym culture makes supplements visible, but no one wants to be seen taking a pill in the library."
3. Finding: In both environments, wellness behaviors that looked cool like fancy water bottles and protein powder were displayed, while "sickness" behaviors i.e. pill bottles were hidden. This social friction prevents adherence in public.
    - Evidence: Students would hunch over or turn away when taking a pill from their bag, whereas they would openly drink from a shaker bottle.

Pros/Cons of this Opportunity Space:

- Pros: This space allows for "wearable" or "portable" IoT solutions, which are exciting to design. There is a "cool factor" to gym/study tech that doesn't exist for home medical devices.
- Cons: Public environments have variable noise and lighting, making sensing difficult. There is also a strong social stigma component as users may not want a device that lights up and announces they need to take a pill in the middle of a quiet library.

Methodology 2: Secondary Research & Social Media Exploration

To supplement our primary observations, the team conducted textual research leading to a Thematic Analysis of Bounded Social Media Places. We focused on digital communities where medication adherence is a frequent topic of discussion, specifically subreddits such as r/ADHD, r/birthcontrol, and r/Biohackers.

Analyzing these digital spaces allowed us to observe candid, peer-to-peer discussions about the "invisible" failures of current adherence tools. Key takeaways included:

- The "Double-Dose" Anxiety: A recurring theme in r/ADHD was the "working memory gap" and users mentioned they frequently could not remember if they had just taken their pill or simply intended to take it, leading to dangerous double-dosing or skipped days.
- Aesthetic Friction: Discussions in r/birthcontrol often centered on the "medicalized" look of blister packs. Users expressed a desire for tools that felt more like "lifestyle accessories" rather than medical reminders, supporting our finding that social concealment is a major barrier.
- Workaround Innovation: We observed users sharing creative "analog" hacks (e.g., flipping a bottle over, using rubber bands) which further validated that current digital app notifications are insufficient and easily ignored.

This thematic analysis provided us with a broader set of "user voices" beyond our local participants, confirming that our identified themes of Visual Noise and Social Stigma are widespread issues in the student-aged demographic.

Instructor/Peer Feedback

The peer feedback we received on our team's project ideas was generally very positive. In the context of healthcare, medication adherence tools and smart dispensers were regarded as very useful, not only for elderly users but also for neurodivergent users, which our team had not previously considered. Our peers suggested considering notifications for both users and caregivers, as well as design considerations for younger users. The Hero app/device was mentioned as a competitor, and a suggestion was made to broaden the audience beyond the

elderly. Other healthcare ideas were praised for their potential use and impact, but our peers recommended defining our scope and examining existing devices. Ideas like a newborn monitor or assistance for new mothers were noted as promising, but could present difficulties when conducting interviews and user research. In the context of smart objects, shopping cart and robot assistance ideas were well-received, and one peer suggested building upon an existing product of “Marty” by Stop&Shop. Peers also had specific ideas for improving our smart shopping cart idea like integrating automatic checkout. Even our more ridiculous ideas like a smart fridge that shames office coworkers for stealing food were well-received, and our peers pointed out that accountability features like that could be adapted to other contexts. The instructor feedback we received was to really consider if the audiences we were aiming for were recruitable, such as new mothers.

The feedback we received from both peers and instructors was extremely helpful to our team. It prompted immediate discussion among ourselves, and we really took the “recruiting for user research/interview” comments seriously. After receiving feedback, we decided that although we were passionate about technologies that could serve new mothers or the elderly, we didn’t have access to potential users in these audiences. From our peer feedback, we realized that neurodivergent users or students that take regular medication would be a feasible audience for any medication adherence ideas we explore. We also became more open as a group to some of our “ridiculous” ideas that we had merely included in the presentation as examples. Based on our peer feedback, we realized the feasibility of some of these ideas. Our peers also pointed out specific competing products we could look into, which was very helpful as a starting point.

Product Scope

Population: College students aged from 18 to 30 who take medication
Environment/Context: We are creating an intervention that will allow users to better adhere to taking their medication, whether at home or on the go
IoT Opportunities:

1) Use connected devices to automatically track self-care progress, transforming these activities into a gamified experience that provides clear, engaging feedback to encourage users to stick to it.
2) Devices that automatically remind students to take medication. Devices may also record medication events through connected systems and upload the data to relevant apps, reducing manual input and cognitive effort.
3) A connected lamp adjusts brightness and color temperature across the day to support waking up, focusing, taking breaks, and winding down, helping students build healthier daily rhythms.

Mini Persona:

1) Persona 1: College student - neurodivergent

    a) Barry (21)(M) takes allergy medication as needed. He often carries his medication on his person and takes it when he knows that he will be exposed to certain environmental allergens. Though Barry may keep his medication on hand, he may sometimes forget it when balancing other life stresses. This was the case recently. At a college event for midterms, Barry was exposed to allergens that caused a reaction. Since he did not have his medication on hand, Barry had to be rushed to the ER.
2) Persona 2: College student - not neurodivergent
    a) Stayce (24)(F) regularly takes medication for anxiety/ADHD. She is quite reliant on this medication and notices that there are significant changes to her symptoms when she regularly takes her medication. However, during high-stress times of the semester, she often becomes so overwhelmed that she forgets to take her medication. This forgetfulness is very detrimental to her academic success, as it sometimes creates a cycle where her symptoms worsen, making it harder for her to focus in class and get tasks done.

Conclusion

Why is the team interested in solving this problem?

Our team is really interested in this concept because it reflects challenges we have personally experienced as college students managing medication in fast-paced, high-stress academic environments. As students balancing coursework, jobs, and social responsibilities, we have firsthand experience with how easy it is to forget or inconsistently take medication, especially during busy or overwhelming periods of the semester. We have also seen friends and peers struggle with medication adherence, including those managing anxiety, ADHD, allergies, or other conditions where missed medication can significantly affect daily functioning and academic performance. These experiences made the problem feel personal and emotionally connected. By focusing our product scope on college students and self-care contexts, we are motivated to explore how IoT solutions can reduce cognitive burden and provide support to the community. Our goal is to design solutions that fit naturally into students' daily routines and ultimately solve their problems or frustrations.

What next steps will the team take in M2?

In Milestone 2, our team will focus on conducting structured user research to deepen our understanding of college students' medication adherence behaviors and challenges. We will begin by preparing a survey and a diary study/cultural probe, which allows us to capture both quantitative and qualitative insights. We will draft our research protocols, participate in peer review, and recruit participants who fit our target population.

Before conducting research, we will review and refine our protocols as a team to ensure the questions are clear, unbiased, and appropriate for real-world contexts. We plan to recruit participants by reaching out to friends and peers who align with our intended audience.

Based on peer feedback, we will iterate on our protocols and conduct research sessions. We will carefully document observations, responses, and artifacts to support effective data collection and analysis. As a team, we will analyze our findings to identify key themes, pain points, and opportunity areas related to medication adherence for college students. We will then synthesize these insights into empathy maps and journey maps, which will inform our ideation of potential design solutions grounded in our research findings.